



Prolonged epileptiform EEG runs are associated with persistent seizures in juvenile myoclonic epilepsy



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ARTICLE INFO

Keywords:

Juvenile myoclonic epilepsy
Spike-wave-duration
Photoparoxysmal
Hyperventilation response
SCORE
Prognosis

ABSTRACT

Objective: In juvenile myoclonic epilepsy (JME), various EEG characteristics have been suggested as poor prognostic signs, but their significance is unclear. The aim of this study was to assess the influence of EEG variables on seizure and psychosocial outcome after a follow-up exceeding 20 years.

Methods: 396 EEG recordings were available for assessment in 40 patients (42 complete digital, 330 paper segments and 24 written reports only). Mean follow-up was 31 years (range 20–68). The number of EEGs per patient ranged from 2 to 23 (mean 9). Twenty-one patients were in remission for > 5 years, whereas 19 had persistent seizures. Favorable psychosocial outcome was found in 14 of 37. EEGs were retrospectively categorized into four main groups; normal, slowing, epileptiform discharges or both slowing and epileptiform discharges, with further sub-classification. Hyperventilation and photoparoxysmal responses were also evaluated. Scoring of EEG findings was blinded to seizure and psychosocial outcome.

Results: Significant associations were found between poor seizure control and prolonged ≥ 3 s epileptiform runs, $p = 0.03$ (8/19 vs 2/21), long ≥ 3 s photoparoxysmal runs, $p = 0.04$ (6/19 vs 1/21) and long ≥ 3 s hyperventilation-induced epileptiform runs, $p = 0.02$ (5/19 vs 0/21). The strongest association between persistent seizures and EEG was found when all epileptiform runs ≥ 3 s were combined ($p = 0.007$), with a positive predictive value equal to 79% and a negative predictive value equal to 69%. Fast (4–5c/s) spike-wave runs were also more frequent in patients with persistent seizures compared to the remission group, $p = 0.04$ (9/19 vs 3/21). Other epileptiform elements occurred equally in the two prognostic groups. Psychosocial outcome was not influenced by EEG findings. Prolonged runs within 6 months from first recording did also predict clinical outcome, $p = 0.03$; (8/19 vs 2/21), with a positive predictive value equal to 80% and a negative predictive value equal to 63%.

Significance: Fast spike-wave runs and prolonged (≥ 3 s) epileptiform runs, including photoparoxysmal and hyperventilation-induced runs were associated with persistent seizures in JME. Focal EEG abnormalities were not associated with clinical outcome. Conceivably, the duration of epileptiform bursts reflects the degree of deficient intracortical inhibition. Prolonged runs may represent an essential predictive feature for poor seizure control in JME.

1. Introduction

Juvenile myoclonic epilepsy (JME) is the most common form of idiopathic generalized (genetic) epilepsy (Syvertsen et al., 2017). Seizure onset usually occurs around puberty with early morning myoclonic jerks. The majority of patients develop generalized tonic-clonic seizures and 20–40% has absence seizures (Commission, 1989; Genton et al., 2013; Janz, 1989; Kasteleijn-Nolst Trenité et al., 2013). EEG is the most useful supplementary diagnostic method. Valproate is

effective in the majority of patients (Gesche et al., 2017; Penry et al., 1989), but reproductive toxicity limits its use in women. Other broad spectrum antiepileptic drugs (AEDs) like levetiracetam, lamotrigine, topiramate and zonisamide are alternatives (Kasteleijn-Nolst Trenité et al., 2013; Syvertsen et al., 2014).

JME has been considered a lifelong disorder, but several recent long-term studies report sustained seizure freedom permitting treatment withdrawal in some patients (Baykan et al., 2008; Camfield and Camfield, 2009; Geithner et al., 2012; Höfler et al., 2014; Kasteleijn-

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Nolst Trenité et al., 2013; Seneviratne et al., 2012; Senf, 2013; Syvertsen et al., 2014). Clinical as well as EEG characteristics predicting the clinical course and the treatment response have accordingly been sought for. However, studies with particular focus on the long-term prognostic value of EEG features are scarce. The number of included patients has been limited and the findings have been diverse. Some authors report an association between EEG focality or asymmetry and poor seizure control (Fernando-Dongas et al., 2000; Jayalakshmi et al., 2010; Szaflarski et al., 2010; Yasemin Biçer et al., 2012), whereas others do not (Betting et al., 2006; Japaridze et al., 2016; Lancman et al., 1994; Letourneau et al., 2010; Senf et al., 2013). Hence, these issues remain to be clarified in larger long-term studies. Moreover, in a pooled material of idiopathic generalized epilepsies, it was recently demonstrated that higher densities and longer durations of epileptiform discharges were associated with shorter seizure free periods (Seneviratne et al., 2017). By reviewing a large number of EEG recordings, we aimed to estimate the associations between selected EEG features based on the recently proposed standardized computer-based organized reporting (SCORE) protocol and the long term seizure outcome of JME. A secondary aim was to explore the association between EEG and psychosocial outcome.

2. Methods

2.1. Patients

The study was based on 42 previously reported patients with JME, originally identified for a study on the association to human leukocyte antigens in 1992 (Moen et al., 1995). In 2012, 40 of these patients were available for seizure outcome assessments 31 years (range 20–68 years) after epilepsy onset (Syvertsen et al., 2014). Median age at follow-up was 47 years (range 35–81). Demographic and clinical data, as well as treatment characteristics are shown in Table 1. The clinical diagnosis of JME was based on the ILAE-classification from 1989 (Commission, 1989). Information on seizure control after at least 20 years of clinical follow-up was available for 40 patients; 21 were in the seizure free (SF) group (in remission for more than 5 years) and 19 were in the persistent seizure (PS) group. Valproate had been used in all, apart from four females (Table 1).

A favorable psychosocial outcome was found in 14 of 37 patients available for interview (Syvertsen et al., 2014).

2.2. EEGs

EEG-reports (by clinical neurophysiologists or neurologists with EEG certification) were identified in all patients. In 39, EEG recordings from the Department of Neurology and Clinical Neurophysiology, St. Olav's Hospital, Trondheim, were available. In four, EEG recordings were retrieved from other hospitals in addition; one patient had only recordings from another hospital. EEG-data on paper or in digital form was available for rescoring in 40 patients. A total of 396 EEGs were available for the study (Table 2). The number of EEGs per patient ranged from 2 to 23 (Table 1). The maximal interval between available recordings was 42 years. The international 10–20 system for electrode placement was used.

EEG recordings were retrospectively reviewed with blinding for clinical outcome. Table 2 shows the various recording types.

2.3. Analyses

The EEGs were categorized into four main groups; normal, focal/regional/generalized slowing, focal/regional/generalized epileptiform discharge or both slowing and epileptiform discharge. Asymmetries, slowing or epileptiform discharge, hyperventilation and photoparoxysmal responses were evaluated. Abnormalities were graded as 1 (1–3 elements per 20 min EEG recording), 2 (4–10 elements) and 3 (> 10

Table 1

Demographic, clinical and treatment characteristics of patients with JME. Mean (sd) and range, or percentage.

	Persistent seizures (n = 19)	Seizure free (n = 21)
Age (yr)	46.4 (7.5) [35–59]	47.9 (9.6) [36–81]
Females	74%	52%
High school graduation	61%	79%
Age at onset (yr)	14.2 (2.9) [8–18]	16.2 (3.9) [6–24]
Follow-up (yr)	31.9 (7.1) [20–43]	31.6 (10.5) [21–68]
Treatment start (age)	15.1 (3.0) [9–20]	18.6 (4.0) [13–26]
Number of EEGs taken	12.2 (5.6) [4–23]	7.1 (4.0) [2–15]
Clinical absences	32%	24%
Myoclonic-tonic-clonic seizures	79%	40%
<i>AEDs at final follow-up</i>		
Polytherapy	47%	33%
Valproate	47%	48%
Levetiracetam	32%	19%
Lamotrigine	21%	19%
Topiramate	16%	14%
Carbazepines ^a	16%	5%
Benzodiazepines ^b	5%	10%
Barbiturates ^c	0%	5%
Zonisamide	0%	5%
<i>AEDs ever used</i>		
Polytherapy	63%	38%
Valproate	89%	90%
Levetiracetam	42%	33%
Lamotrigine	47%	33%
Carbazepines ^a	63%	52%
Topiramate	26%	29%
Benzodiazepines ^b	26%	29%
Barbiturates ^c	37%	14%
Phenytoin	32%	24%
Ethosuximide	16%	10%
Zonisamide	0%	5%

All AED comparisons (Fisher exact test): $p > 0.15$.

^a Carbamazepine/oxcarbazepine.

^b Clobazam/clonazepam.

^c Primidone/phenobarbital.

Table 2

EEG recording types in 40 patients with JME followed for > 20 years.

	No of EEGs	No of patients	Persistent seizures (n = 19)	Seizure free (n = 21)
Total	396	40		
Standard recordings	383	40	226(59%)	157(41%)
Sleep deprived recordings	9	9	4(40%)	5(60%)
Long term monitoring ^a	4	3	2(50%)	2(50%)
Available digital recordings ^b	42	22	26(62%)	16(38%)
Selected segments of paper recordings	330	40	204(62%)	126(38%)
EEG report without preserved recording ^c	24	5	1(4%)	23(96%)

^a Mostly 24 hour ambulatory recording.

^b Including pruned EEG-files.

^c All patients had at least one preserved EEG recording in addition to recordings where only written EEG-reports had been kept.

elements), and subdivided into focal-regional or generalized. Localization was categorized as right or left fronto-central, temporal, or occipito-parietal. Amplitude asymmetry was defined as a consistent > 50% difference.

The morphology of epileptiform discharges was sub-classified based on the definition suggested by the SCORE group in their Appendix S4: Polyspike-and slow waves, polyspikes, rapid spike runs, sharp-waves, sharp-and-slow-wave complex, spike and spike-and-slow-wave. Spike/

polyspike-and slow-wave runs were categorized into classical ($3\text{--}3.5\text{ s}^{-1}$), slow ($1\text{--}2.5\text{ s}^{-1}$), fast ($4\text{--}5\text{ s}^{-1}$) and atypical (Beniczky et al., 2013).

Prolonged epileptiform runs were defined as generalized epileptiform activity lasting $\geq 3\text{ s}$, as this duration is supposed to significantly impact cognitive function (Aldenkamp et al., 2005). The number of prolonged epileptiform runs were counted and their durations were measured. In addition epileptiform discharges $\geq 3\text{ s}$ during photic stimulation and hyperventilation were counted and measured.

Due to a different number of EEGs available for each patient, *relative occurrence variables* were also calculated as the sum of EEGs with each abnormal feature divided by the total number of EEGs recorded in each patient. Main EEG variables were selected based on previous reports regarding the prognostic significance of EEG in JME: 1) Number of patients with at least one abnormal EEG feature (severe or moderate slowing, severe or moderate epileptiform activity, photoparoxysmal response, epileptiform activity triggered by hyperventilation and focality/asymmetry, 2) Relative occurrence of main EEG features among all EEGs within each subjects, expressed statistically as mean percentage (and SD) for patients in SF and PS groups, and 3) numerical data averaged within patients (alpha frequency, duration of epileptiform runs in seconds).

EEG features were retrospectively summarized across the entire follow-up period. In addition, we also summarized EEG variables for the first recordings in each patient in order to explore the prognostic significance of the first EEGs.

Data analysis was performed using SPSS version 21. Data were characterized by using descriptive statistics (mean and SD or frequencies and percentage when appropriate). EEG features in the two prognostic patient groups (and the two psychosocial outcome groups) were compared using Fisher exact test for categorical 2×2 data, while the non-parametric Mann–Whitney test was used to compare numerical data (e.g. *relative occurrence* variables) between groups. Differences in proportions within groups were evaluated with a binomial test. Larger contingency tables were evaluated with a chi-square test. A p -value < 0.05 was considered significant.

The study was approved by the Regional Committee for Medical Research Ethics, and written informed consent was provided by the participants.

3. Results

3.1. EEG graphoelements

The main EEG findings are summarized in Tables 3–6. Fast spike-and-slow-wave runs and long ($\geq 3\text{ s}$) epileptiform runs were more frequent in the PS group (Table 3). Regular spike-and-wave activity (excluding poly-spike waves) also tended to occur more frequently in the PS group. The occurrence of other specific epileptiform features was equal in both prognostic groups (Table 3).

A statistically significant association was seen with poor seizure control and the occurrence of photoparoxysmal runs $\geq 3\text{ s}$ and epileptiform runs $\geq 3\text{ s}$ during hyperventilation both for occurrence (Table 4) and the relative frequency of abnormal EEGs (Table 5). However, brief photo-paroxysms and epileptiform hyperventilation response rates were equal in the clinical outcome groups (Table 4).

The strongest association between PS and EEG was found when all epileptiform runs $\geq 3\text{ s}$ were combined ($p = 0.007$; Table 4), with a positive predictive value equal to 79% and a negative predictive value equal to 69%. This pattern was documented in altogether 37 recordings in 14 JME patients; in the first EEG for seven of them and within the first six months in another three. Prolonged epileptiform runs within six months from the first EEG were also associated with PS ($p = 0.03$; Table 4) with a positive predictive value equal to 80% and a negative predictive value equal to 63%. However, EEG features in the very first recordings showed no significant differences between the seizure

Table 3

Patients with either slowing or unprovoked epileptiform abnormalities in at least one EEG recording.

	Persistent seizures ($n = 19$)		Seizure free ($n = 21$)		p -value ^a
	n	%	n	%	
Any diffuse or focal slowing	16	(84)	17	(81)	ns
Moderate or severe diffuse or focal slowing ^b	9	(47)	6	(29)	ns
Any epileptiform activity	18	(95)	20	(95)	ns
Patients with moderate or severe epileptiform EEG activity ^b	14	(74)	12	(57)	ns
Polyspike-waves ^b	10	(53)	9	(43)	ns
Polyspikes ^b	2	(11)	3	(14)	ns
Spikes ^b	6	(32)	3	(14)	ns
Spike-wave activity ^b	12	(63)	6	(29)	(0.06)
Any epileptiform SCORE run type (classic/fast/slow/atypical) ^c	12	(63)	11	(52)	ns
Fast (4–5 c/s) spike-and-slow-wave runs	9	(47)	3	(14)	0.04
Patients with ≥ 3 second spontaneous epileptiform runs	8	(42)	2	(10)	0.03

^a Fisher exact test (two-sided).

^b Excluding slight and uncertain findings.

^c Run-duration is not a part of the SCORE-classification.

ns, $p > 0.07$.

prognostic groups.

We found no statistically significant association between the prognostic groups and the occurrence of normal EEG findings within subjects, the combination of all epileptiform discharge types, abnormal background activity or focal slowing (Table 5). No association was noted between alpha frequency (data not shown) or focal/asymmetric EEG features and seizure outcome (Table 6).

No association between EEG features and psychosocial outcome was seen (data not tabulated).

3.2. Seizure types

Clinically reported seizure types related to prolonged runs are shown in Table 7. Absences were associated with prolonged runs in 7/14 patients (50%), whereas absences had been reported in 4/26 patients with short runs only (15%) ($p = 0.02$). Other seizure types were not related to prolonged runs.

3.3. Antiepileptic drugs

Of the 19 patients using valproate at the end of the study (10 SF/9 PS), eight had prolonged runs. Seven of these eight patients had persistent seizures and only one was seizure free ($p = 0.035$; binomial probability test), corroborating the association of prolonged EEG runs with pharmacoresistance.

Among the 178 recordings in 14 patients with at least one prolonged run valproate was used during 79 recordings in 12 patients, whereas carbamazepine was used in 58 recordings in 11 patients, levetiracetam was used in 3 recordings in 2 patients, lamotrigine in 5 recordings in 3 patients and topiramate in 2 recordings in one patient (Table 8). The distribution of patients across AEDs were similar in patients with and without prolonged runs ($p = 0.77$, chi-square test with 9 degrees of freedom). Patients ever exhibiting prolonged runs had EEGs performed at slightly younger mean age (Table 8), but the difference was not significant ($p = 0.36$). However, when age at recording was compared between 37 EEGs with runs (mean 19.9 years) and all other EEGs (mean = 24.9 years) the age-difference was larger. Only 11 EEGs with

Table 4

JME patients with epileptiform discharges during either photostimulation or hyperventilation in at least one EEG recording.

	Persistent seizures (PS group, <i>n</i> = 19)		Seizure free (SF group, <i>n</i> = 21)		<i>p</i> -value ^a
	<i>n</i>	%	<i>n</i>	%	
Any photoparoxysmal epileptiform discharge	15	(78)	14	(67)	ns
Moderate/severe photoparoxysmal epileptiform discharge ^b	12	(63)	12	57	ns
Photoparoxysmal epileptiform runs ≥ 3 s	6	(32)	1	(5)	0.04
Any epileptiform activity during hyperventilation	14	(74)	10	(48)	ns
Moderate/severe epileptiform activity during hyperventilation ^b	8	(42)	5	(24)	ns
Epileptiform runs ≥ 3 s during hyperventilation	5	(26)	0	(0)	0.02
Any epileptiform run ≥ 3 s (HV, photic or spontaneous)	11	(58)	3	(14)	0.007
Any epileptiform run ≥ 3 s (HV, photic or spontaneous) within first 6 months	8	(42)	2	(10)	0.03

^a Fisher exact test (two-sided).^b Abnormality defined as either moderate or severe, excluding slight and uncertain findings; ns, *p* > 0.05.**Table 5**

Relative frequency of EEG findings and lateralized EEG findings (% of all EEGs within subjects).

	Persistent seizures (<i>n</i> = 19)		Seizure free (<i>n</i> = 21)		<i>p</i> -value ^a
	Mean	SD	Mean	SD	
Normal (%)	27	(29)	36	(25)	ns
Epileptiform discharges (%)	54	(29)	45	(30)	ns
Generalized and focal slowing (%)	19	(18)	19	(20)	ns
Focal slowing (%)	12	(19)	10	(12)	ns
Photoparoxysmal runs ≥ 3 s (%)	7	(12)	2	(8)	0.04
Epileptiform runs ≥ 3 s during hyperventilation (%)	4	(7)	0	(0)	0.01
Focal slowing right hemisphere (%)	17	(34)	2	(9.7)	(0.06)
Focal slowing left hemisphere (%)	10	(17)	12	(33)	ns
Focal epileptiform discharge right hemisphere (%)	7	(13)	7	(16)	ns
Focal epileptiform discharge left hemisphere (%)	6	(14)	6	(13)	ns

^a Mann-Whitney test (asymptotic significance); ns *p* > 0.05.**Table 6**JME patients with epileptiform focality or asymmetry^a in at least one EEG recording.

	Persistent seizures (<i>n</i> = 19)		Seizure free (<i>n</i> = 21)		<i>p</i> -value ^b
	<i>n</i>	%	<i>n</i>	%	
Frontocentral epileptiform activity	6	(32)	5	(24)	ns
Temporal epileptiform activity	6	(32)	4	(19)	ns
Parietooccipital epileptiform activity	3	(16)	2	(9)	ns
Asymmetric epileptiform activity ^a	15	(79)	18	(86)	ns
Asymmetric epileptiform runs ^a	7	(37)	6	(29)	ns
Asymmetric focal/regional epileptiform activity ^a	10	(53)	10	(48)	ns
Asymmetric generalized epileptiform activity ^a	13	(68)	13	(62)	ns

^a Amplitude difference > 50% in focal/regional or generalized runs.^b Fisher exact test (two-sided); ns, *p* > 0.05.**Table 7**Seizure types in JME patients with or without prolonged (≥ 3 s) runs [*n*(%)].

	Patients without any prolonged EEG-runs	Patients with at least one prolonged EEG-run
Number of patients	26	14
Females	12	13
Age at debut (yr)	15.6 (6–24) ^a	14.6 (9–20) ^a
<i>Seizure distribution</i>		
Myoclonic only	1	0
Myoclonic tonic-clonic	15	9
Tonic-clonic	24	14
Clinical absence	4	7
Persistent seizure	8	11

^a mean(range).**Table 8**Treatment (AEDs at recording) and diurnal characteristics associated with recordings with or without prolonged (≥ 3 s) runs [*n*(%)].

	Recordings from patients without prolonged runs (Number of patients)	Recordings from patients with at least one prolonged run ^a (Number of patients)
Mean age at recording time (yr)	26.1	22.7
Number of recordings (patients)	218 (26)	178 (14)
Barbiturates ^b	29 (6)	14 (3)
Benzodiazepines ^c	4 (3)	15 (3)
Carbamazepine ^d	24 (9)	58 (11)
Ethosuximide	9 (2)	8 (1)
Lamotrigine	19 (7)	5 (3)
Levetiracetam	11 (1)	3 (2)
Phenytoin	20 (5)	30 (4)
Topiramate	12 (5)	2 (1)
Valproate	104 (18)	79 (12)
Zonisamide	1 (1)	0 (0)
Prior to treatment	26 (18)	11 (8)
After discontinuation	16 (7)	0 (0)
<i>Time of day for EEG and sleep-EEGs</i>		
Early recording (< 11:30 AM)	58 (22)	53 (13)
Late recording (> 11:30 AM)	81 (23)	55 (13)
Unknown	79 (16)	70 (11)
Sleep deprived recordings ^e	6 (6)	3 (3)

AED, antiepileptic drug.

^a 37 of 178 recordings showed prolonged runs.^b Phenobarbital/primidone.^c Clobazam/clonazepam.^d Carbamazepine/oxcarbazepine.^e Prolonged runs were not recorded during sleep.

prolonged runs were performed without AEDs (11 prior to treatment and none after discontinuation) (Table 8).

3.4. Diurnal variation of EEG recording

Exact time of the day was known for 108 recordings in patients ever showing prolonged bursts; 53 recordings in 13 patients (30%) were performed at early hours (before 11:30 AM), while 55 recordings in 13 patients were recorded later (31%). In these EEGs, prolonged runs were identified in 18; nine (50%) were recorded during morning sessions. Prolonged runs were not found in any sleep-recording; only nine of the EEGs were performed after sleep deprivation (Table 8).

4. Discussion

4.1. Prolonged epileptiform runs

The most important findings in the present study is that long-lasting (≥ 3 s) epileptiform runs, either unprovoked, during photic stimulation or hyperventilation, were associated with persistent seizures in JME. Although it is well known that spike-wave runs is a common EEG feature in JME (Shahnaz et al., 2014), the duration of such runs at rest, or during photic stimulation and hyperventilation, has not received much attention in the literature. Spike and poly-spike wave runs ≥ 3 s seem to be rather common in JME patients with a history of absences, although clinical symptoms during discharges often are subtle or lacking (Panayiotopoulos et al., 1989). Absences have been associated with poorer prognosis in JME in some previous studies (Höfler et al., 2014; Senf et al., 2013). In the present study absences were also associated with prolonged runs, suggesting that prolonged runs are closely related to the pathophysiological processes determining prognosis as well as clinical absences.

To the best of our knowledge, the prognostic significance of such runs is a novel observation. This is in line with a recent Australian study on idiopathic generalized epilepsies (JME and epilepsy with GTCs alone, and absence epilepsies). The duration of generalized paroxysms in 24 h recordings was strongly associated with the duration of preceding seizure free periods. Those with longer paroxysms were more likely to have had recent seizures (Seneviratne et al., 2017). Accordingly, the duration of epileptiform bursts may reflect the degree of deficient intracortical inhibition within the epileptogenic network, thus rendering prolonged spiking as an essential predictive feature for persistent seizures in JME (Badawy et al., 2010; Serafini et al., 2013).

4.2. Generalized epileptiform activity

Interictal EEG findings in JME are classically described as generalized bilateral and symmetric rapid (4–5 c/s) spike-waves and polyspike-waves with normal background activity (Commission, 1989; Janz, 1989; Serafini et al., 2013). This was also confirmed in the present study. In addition, fast 4–5 c/s runs were associated with persistent seizures. However, no association with seizure persistence was found for epileptiform features in general, for instance polyspike-waves, in concert with $\geq$ other studies on seizure frequency (Baykan et al., 2008; Senf et al., 2013). Another group reported an association between a favorable seizure outcome and occurrence of normal EEGs or symmetric generalized spike-waves in idiopathic generalized epilepsies as a whole (Szaflarski et al., 2010).

4.3. Response to activation procedures

Epileptiform discharges are commonly provoked by photic stimulation and hyperventilation in JME (Kasteleijn-Nolst Trenite et al., 2013). A photoparoxysmal response has been reported in about one third of JME patients (Appleton et al., 2000; Jayalakshmi et al., 2010) or up to 90% with prolonged stimulation (Appleton et al., 2000). Some studies

report no association between a photoparoxysmal response and seizure frequency (Baykan et al., 2008; Senf et al., 2013), whereas one study showed that it significantly increased the risk of seizure recurrence after AED discontinuation (Geithner et al., 2012).

We confirm that photoparoxysmal responses are frequent in JME, and in line with most of the literature, we found no association between the most common short-lasting runs and seizure persistence. However prolonged photoparoxysmal runs ≥ 3 s occurred more frequently in the PS than the SF group. Similarly, long-lasting fast epileptiform runs during hyperventilation were observed more frequently in the PS group compared to the SF group, suggesting that provoked and spontaneous runs reflect rather identical intracortical epileptogenic mechanisms. Since short-lasting epileptiform excitatory elements like spikes and polyspike-waves were not associated with prognosis, our findings corroborate the role of a defective intracortical or cortico-cortical inhibition (Badawy et al., 2010; Serafini et al., 2013).

4.4. Focal abnormalities and asymmetries

EEG asymmetries have been found in 12–54% of JME patients in previous studies (Aliberti et al., 1994; Leutmezer et al., 2002; Lombroso, 1997; Usui et al., 2005; Yasemin Biçer et al., 2012). Apparent focal EEG abnormalities can occasionally be observed and may lead to misdiagnosis and inappropriate treatment. Several studies have reported an association between seizure control and focal EEG abnormalities, either as non-specific slowing, epileptiform discharges or both (Jayalakshmi et al., 2010; Yasemin Biçer et al., 2012). Focal abnormalities in the first recording was a predictor of poor response to valproate treatment in one study (Szaflarski et al., 2010), and focal EEG features have been found in as many as 30–35% of patients with idiopathic generalized epilepsies with poorly controlled seizures (Leutmezer et al., 2002). Another study found that EEG asymmetries were associated with valproate resistance (Fernando-Dongas et al., 2000). However, the association between focality/asymmetry and prognosis is still somewhat controversial due to inconsistent findings in several previous JME studies (Betting et al., 2006; Lancman et al., 1994; Letourneau et al., 2010; Senf et al., 2013). This discrepancy might be explained by methodological differences, because some refer to amplitude asymmetries as a focal or asymmetric EEG feature, some define the amplitude asymmetry as $> 30\%$ (Szaflarski et al., 2010), others as discharges with $> 50\%$ voltage asymmetries observed in more than one EEG (Lancman et al., 1994), while one study defines EEG asymmetries as difference of amplitude of generalized discharges $> 50\%$ or focal onset preceding the generalized discharges (Jayalakshmi et al., 2010). Localized EEG findings were not associated with seizure outcome in the present study, supporting that temporal features like run-duration are more important than spatial features like network activation asymmetry.

4.5. Long-term significance of EEG findings

Previous long-term follow-up studies of patients with JME have assessed long-term EEG findings to a variable extent. A German study evaluated the first EEGs for spike-waves appearance, photosensitivity, focal abnormalities or abnormal background activity, but subsequent EEGs were not assessed (Senf et al., 2013). Although the results of the first EEG did not carry significant prognostic weight, we observed that our strongest predictor, combined epileptiform runs ≥ 3 s, was present within 6 months in 10 of 14 with this pattern. Conceivably, the results from more than one EEG summarized over about 6 months are needed to demonstrate the prognostic importance of prolonged epileptiform runs.

4.6. Psychosocial outcome

Previous studies have reported a high risk of unfavorable psycho-

social outcome in JME (Camfield and Camfield, 2009; Syvertsen et al., 2014). Unsurprisingly, our findings suggest that psychosocial outcome is too complex and multidimensional to be prognosticated by serial EEGs.

4.7. Limitations and strengths of the study

Our study is retrospective and spanning over several decades. We have considered various potential confounders, such as the influence of AED treatment effects and the diurnal variation of cortical excitability in JME. Variable numbers and types of EEG recordings were available for each patient. The number of patients included was limited, but the proportion of patients with seizures within the last 5 years was relatively high, conceivably due to the tertiary function of the clinic. However, seizure breakthrough in JME is often the result of irregular lifestyle, such as treatment non-adherence, sleep loss and alcohol intake (Sokic et al., 2007). Elements of such “pseudo drug resistance” could not be controlled for.

In addition, some of the patients were not drug naïve (11/40) at the time of the first available EEG recording. The PS group was subject to a more intense follow-up than the SF group with a higher number of EEGs, including long-term recordings, as well as more treatment change and polytherapy (Table 1), a natural consequence of persistent seizures.

Broad-spectrum AEDs, appropriate for use in generalized epilepsies, are known to inhibit epileptiform EEG activity (Clemens, 2008; Eriksson et al., 2001; Specchio et al., 2008; Sundqvist et al., 1999). Medication effects could not be fully controlled for in this long-term retrospective study with highly variable EEG-recordings per patient. However, AED-use was not different among patients with and without long runs. Nevertheless, nearly all had been exposed to valproate (apart from four females). At final follow-up, valproate use was equally distributed in the two prognostic groups; it is thus unlikely that the present results are significantly influenced by the effect of this drug. Moreover, the use of other AEDs was also largely similar apart from a trend to more use of newer AEDs at follow-up in the PS group.

In JME, the occurrence of epileptiform discharges is increased after awakening and at sleep onset (Labate et al., 2007; Serafini et al., 2013). In the present study, the exact time of the recordings was known for only a limited number of EEGs. The distribution of recordings in patients with prolonged runs at any time was equal in early and late recording sessions. Association with sleep could not be studied due to the low number of sleep-recordings.

We did not find consensus definitions for focal features, and we defined EEG-focality based on descriptions in previous studies. However, for EEGs with limited paper-based selections available, it is possible that some focal EEG features and some prolonged runs may have been missed.

Since we intended to control for type 2 errors, and since we only compared two prognostic groups, Bonferroni correction was not applied (Perneger, 1998). However, a type 1 error can never be excluded for any significant result, particularly when several outcome-variables are assessed. Hence, independent replication will always be necessary. It is reassuring that a very recent study, supports our main finding (Seneviratne et al., 2017)

One strength of the present study is that morphology of epileptiform discharges was classified by the proposed consensus of the SCORE working group from 2013. Interrater blinded agreement was thus not considered necessary. Other strengths are the rather long study period and the relatively high number of EEGs. Most patients resided in the primary catchment area of the hospital, Sør-Trøndelag county, with only one neurological and EEG department; thus facilitating this form of long-term follow-up. In addition, the EEG evaluations were blinded for clinical outcome and treatment.

4.8. Future directions and conclusions

The discrepancies between many previous studies may be due to the fact that different definitions and methods have been used. For instance, the definition of focal EEG features have varied considerably, and it would be helpful if objective scoring criteria for asymmetry were defined by international consensus, e.g. in SCORE, to be used in future studies. We also recommend that epileptiform run/burst durations should be included in future revisions of the SCORE system since the present study, for the first time, indicates that prolonged ≥ 3 s spontaneous and stimulation-induced epileptiform runs are associated with persistent seizures in JME. We suggest that these findings may correlate to the degree of deficient intracortical inhibition, which may be one essential pathophysiological feature linked to treatment resistance in JME. New EEG-protocols with prolonged activation procedures may possibly also be useful for detecting features that reflect decreased cortical inhibition and poor seizure control. Future prospective studies with structured protocols taking potential confounding factors into account, should seek to confirm that fast spike-wave runs and prolonged epileptiform runs predict the course of JME.

Conflict of interest

Eylert Brodtkorb has received speaker and/or advisory board honoraria from Eisai, GlaxoSmithKline, UBC and Novartis.

The other authors have no disclosures.

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